

# OpeNTF2: Fairness-aware Graph Neural Team Formation

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## Abstract

We contribute OpeNTF2, a unified framework for neural team formation, which achieves state-of-the-art efficacy while improving efficiency in recommending teams of experts who are *almost surely* successful at completing complex tasks. OpeNTF2 extends its initial release in four prime directions: it (1) integrates fairness-aware debiasing rerankers to mitigate popularity and gender disparities; (2) models team formation as link prediction on expert-skill collaboration graphs to capture multi-hop cross-team relational dependencies via graph neural networks; (3) formulates team formation as a sequence prediction task to leverage encoder-decoder recurrent models and transformers; and (4) introduces a temporal training strategy to account for the evolution of experts' skills and collaboration ties. While prior libraries support reproducible pipelines, none incorporate fairness, sequential and graph-based modeling, and temporal dynamics within a single framework. The codebase, along with the installation and execution instructions as well as case studies of extension components, can be obtained under cc-by-nc-sa-4.0 license at <https://github.com/fani-lab/OpeNTF/tree/v0.2.0.0>.

## CCS Concepts

• Information systems → Recommender systems; • Human-centered computing → Social recommendation; • Computing methodologies → Neural networks .

## Keywords

Fair Team Formation; Social Information Retrieval; OpeNTF;

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\*Fair team formation component.

†Graph neural team formation component.

‡Translative team formation component.

§Component-based package installation.

¶Temporal team formation component, architecture, and continuous integration.



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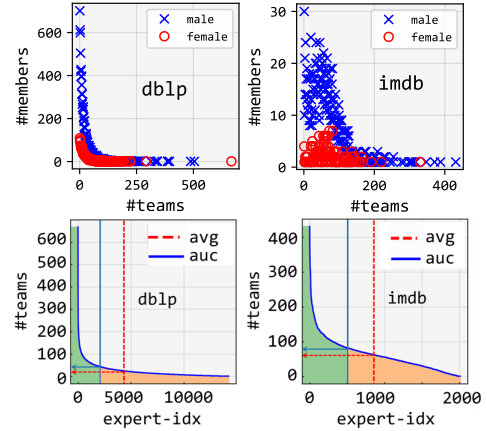


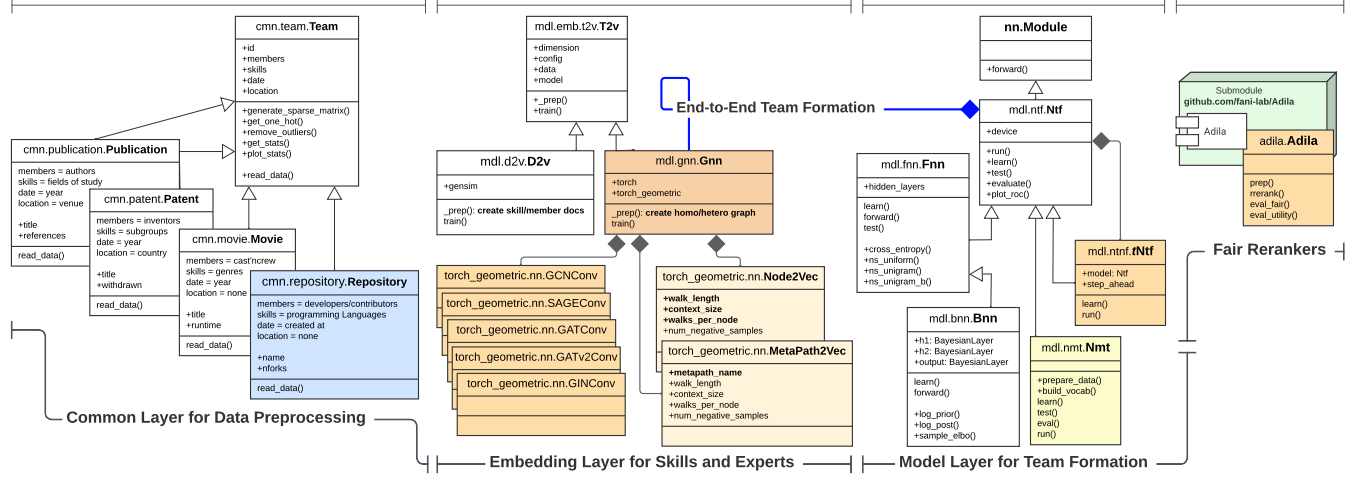
Figure 1: Gender disparity (top) and long-tail distribution of nonpopular experts (bottom) in dblp and imdb [39, 40].

## 1 Introduction

Team formation aims to recommend experts whose combined skills, applied in coordinated teamwork, successfully solve difficult tasks, and can be seen as social information retrieval (Social IR) where the right group of experts are searched for and hired to solve the task at hand. Examples are forming a team of cast and crew to produce a *blockbuster* movie in the genres of ‘action’ and ‘comedy,’ or a team of researchers in the fields of ‘biomedical’ and ‘machine learning’ whose findings are yet to be *published* in a reputable journal. Forming teams is challenging for the large pool of candidate experts from diverse backgrounds and personality traits, along with the *unknown* synergistic balance among them; *not* all teams with the best experts are successful [24, 64]. To address the complex nature of team formation, machine learning has been employed, among which neural models build up the following in scalability and recommendation efficacy [6, 42, 43, 46, 53]. To support neural team formation research with a reproducible and open-source platform, Rad et al. released PyTFL [46], a python-based library that uses a variational multi-label neural classifier to predict the optimal subset of experts for a team. PyTFL, however, struggles with large-scale datasets and lacks modularity for ease of customization and extension to new models and training datasets. To fill the gap, we previously open-sourced OpeNTF [7], a modular and scalable benchmark framework, which includes reference neural models

**Table 1: Comparison of existing libraries for neural team formation.**

|            | neural models                                                        | datasets                    | training strategies                                  | fairness-aware                                             | notions of fairness                                |
|------------|----------------------------------------------------------------------|-----------------------------|------------------------------------------------------|------------------------------------------------------------|----------------------------------------------------|
| PyTFL [46] | variational                                                          | dblp                        | shuffled                                             | ×                                                          | ×                                                  |
| Adila [39] | ×                                                                    | ×                           | ×                                                    | greedy deterministic [16]                                  | demographic parity [11]                            |
| OpeNTF [7] | feedforward, variational                                             | dblp, imdb, uspt            | shuffled, negative sampling                          | ×                                                          | ×                                                  |
|            | feedforward, variational                                             |                             |                                                      |                                                            |                                                    |
| OpeNTF2    | +gnn (transfer)<br>+gnn (end-to-end)<br>+seq-to-seq<br>+transformers | dblp, imdb, uspt<br>+github | shuffled, negative sampling<br>+temporal (streaming) | greedy deterministic [16]<br>+probabilistic reranking [63] | demographic parity [11]<br>+equal opportunity [23] |

**Figure 2: OpeNTF2 partial class diagram. Coloured (gray) components are extensions to our previous release [7].**

along with negative sampling heuristics [6] and three large-scale datasets from varying domains.

Nonetheless, the focus of these and existing libraries has been the maximization of the models' efficacy, largely ignoring fairness, especially when training datasets withhold biases. From Figure 1 (top), datasets in team formation are segmented toward male experts, and females are heavily underrepresented. From Figure 1 (bottom), such datasets also suffer from popularity bias, where the majority of *nonpopular* experts have scarcely participated in teams, whereas few popular experts dominate. Therefore, popular or male experts would receive more attention and are more frequently recommended by neural models, leading to systematic discrimination against already disadvantaged nonpopular or female experts. To the best of our knowledge, there is no fairness-aware library in neural team formation except that of Adila [39], which has adapted deterministic greedy reranking algorithms [16] to mitigate popularity bias. However, Adila is *not* an end-to-end pipeline; it only operates on the recommendation outputs of other libraries. Moreover, existing libraries frame the team formation problem as a *multi-label* Boolean classification based on the mutually independent selection of experts, overlooking both team-level joint dependencies and multi-hop cross-team relational structure. They also learn vector representations of experts and skills in a *static* latent space, failing to capture their temporal dynamics due to societal demands, novel

technologies, and working experiences; successful collaboration of experts years ago would *not* tailor a successful one in the future.

**Contributions.** As summarized in Table 1 and from Figure 2, we bring forth OpeNTF2 that extends the prior version and contrasts with existing libraries in four major directions:

(1) **Fair Team Formation** [16, 39, 63]. To counter unfair biases in the recommended experts for teams by neural models, we adopt probabilistic [63] as well as deterministic [16] debiasing reranking methods based on two well-known alternative notions of fairness: *i)* demographic parity [11] and *ii)* equal opportunity [23], which enables further post-processing refinements to the list of recommended experts to ensure the desired fair outcome in terms of popularity or gender.

(2) **Graph Neural Team Formation** [1, 32, 43]. To capture multi-hop relational and structural information encoded in the expert collaboration network, we integrate graph neural networks in two ways: *i)* prior pretraining and transfer of dense vector representations of skills to accommodate a large set of skills and reduce the complexity of the neural classifiers' *input* layer, following existing work [32, 42, 43]; and *ii)* *end-to-end* team formation via link prediction between required skills and optimal experts for a team, performed directly on the expert collaboration graph, following the efficacy of end-to-end graph neural networks in various recommendation and information retrieval tasks [4, 14, 25, 27, 33, 37, 56, 57, 68].

(3) **Translative Team Formation** [53]. To accommodate the large number of experts for neural classifiers in the *output* layer, we reformulate the team formation problem into a translation task from a sequence of required skills to a sequence of optimum experts and incorporate transformers and encoder-decoder recurrent models with attention for team formation, following their efficacy not only in natural language tasks, but also in recommendation systems [8, 30, 51].

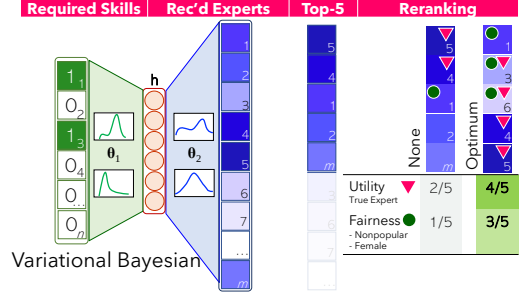
(4) **Temporal Training Strategy** [12]. Inspired by curriculum learning [58] and temporal latent space inference [66, 67], we develop a temporal training strategy that neural models can seamlessly utilize during training to learn temporal aspects of experts' skill sets, levels of expertise, and collaboration ties through the succession of time intervals and to boost the efficacy of inference [12]. **Enhancements.** To further enrich the testbed, we add new evaluation metrics, including skill coverage [44, 55] to capture the coverage of required skills in the recommended team, and a new large-scale *github* dataset, where repositories correspond to teams, programmers to experts, and programming languages to skills. Contrary to existing datasets, *github* has a limited variety of skills, opening new challenges for the generalizability of team formation models in varying domains with distinct distributions. To enhance the architecture, we transition to component-based design in OpeNTF2, essential for integrating a variety of models and training strategies in a unified framework; components are loosely coupled and reusable, dependencies are installed on demand at the component level for execution efficiency and prevent package conflicts, and configurations are managed by *hydra* [60], enabling systematic and extensive ablation studies. OpeNTF2 further incorporates continuous integration and testing along with containerized execution for consistent and reproducible behavior across environments.

## 2 Neural Team Formation

Given a set of skills  $\mathcal{S}$  and a set of experts  $\mathcal{E}$ , a team of experts  $\mathbf{e} \subseteq \mathcal{E}$  that collectively cover the skill set  $\mathbf{s} \subseteq \mathcal{S}$  is shown by  $(\mathbf{s}, \mathbf{e})$  along with its success status  $y \in \{0, 1\}$ , and  $\mathcal{T} = \{(\mathbf{s}, \mathbf{e})_y : y \in \{0, 1\}\}$  indexes all teams. Examples of teams include *dblp* research papers, where authors serve as the experts and fields of study (keywords) act as the associated skills. Similarly, *imdb* movies can be seen as teams comprising cast and crew members as the experts, while genres and subgenres serve as the skills. The choice of movies in team formation literature is *not* to be confused with its use cases in item recommender systems or review analysis; herein, the goal is to form a team of cast and crew for a *movie production*, rather than to recommend movies.

Given a subset of required skills  $\mathbf{s}$  for an *unseen* team, the team formation problem aims to recommend an optimal subset of experts  $\mathbf{e}$  such that their collaboration is *almost surely* successful, that is  $(\mathbf{s}, \mathbf{e})_{y=1}$ , while avoiding subsets of experts  $\mathbf{e}'$  that would result in  $(\mathbf{s}, \mathbf{e}')_{y=0}$ . To this end, neural team formation models learn a mapping function  $f_\theta : \mathcal{S} \rightarrow \mathcal{E}$ , parameterized by  $\theta$ , from  $\mathcal{T}$  using a neural network by maximizing the posterior probability of  $\theta$  such that  $f_\theta(\mathbf{s}) = \mathbf{e}$ , that is,

$$\operatorname{argmax}_\theta p(\theta|\mathcal{T}) \propto p(\mathcal{T}|\theta)p(\theta) = p(\theta) \prod_{(\mathbf{s}, \mathbf{e})_{y=1} \in \mathcal{T}} p(\mathbf{e}|\mathbf{s}, \theta) \quad (1)$$



**Figure 3: Post-hoc debiasing reranking in OpeNTF2. Although illustrated for a variational feedforward model, the reranking is model-agnostic and applicable to graph-based, translative, and temporal models [39, 40].**

where  $p(\mathcal{T}|\theta)$  is the likelihood and  $p(\theta)$  is the prior joint probability of parameters, which is unknown but can be estimated as Gaussian distributions by a variational Bayesian neural network (Bnn) [43–45]. Alternatively, we can assume a *uniform* probability distribution over all possible real-values of  $\theta$  and only estimate the likelihood  $p(\mathcal{T}|\theta)$  via maximum likelihood optimization by a *non-variational* feedforward neural network (Fnn), discarding prior uncertainty  $p(\theta)$  at the cost of overconfident point estimates of  $\theta$  [19]. Non-variational neural networks are primarily used as baselines to demonstrate the performance advantages of variational neural networks [6].

## 3 OpeNTF2 Extensions

Herein, we lay out the details of OpeNTF2's key extensions, along with a brief overview of benchmark results.

### 3.1 Fair Team Formation

From Figure 3, a trained neural model, whether feedforward [6, 45], graph-based [32, 43], translative [53], or temporal [12], recommends *all* available experts but with different probabilities after a sigmoid layer, and the final recommended team includes experts with top- $k$  highest probabilities. This enables further post-hoc *reranking* of the list of recommended experts to ensure the desired fair and unbiased distribution of experts. Here, we provide a brief and high-level overview of the principal components; for theoretical depth and details, we refer to our fairness-related papers [39, 40], also linked in the codebase. For sample runs following best practices, see our colab script at the codebase.<sup>1</sup>

**3.1.1 Debiasing Rerankings.** OpeNTF2 integrated debiasing rerankers including three deterministic [16] and the state-of-the-art probabilistic (*fa-ir*) [63] ones. For instance, in *fa-ir*, the top- $k$  ranked list of a model's prediction is assumed to follow  $k$  independent Bernoulli trials of win/lose at achieving the desired fair team. Given  $p$  as the desired proportion of disadvantaged experts (nonpopular or female) in a fair team and a significance level  $\alpha$ , at each position  $i$ , we check whether top- $i \in \{1, \dots, k\}$  ranked list statistically significantly follows a Bernoulli distribution with winning probability  $p$ .

<sup>1</sup><https://github.com/fani-lab/OpeNTF/blob/main/ipynb/fair.ipynb>

Otherwise, based on a *notion* of fairness, we select a disadvantaged expert from the list after  $i$  and insert her at the  $i$ th position.

**3.1.2 Notions of Fairness.** To design and evaluate fairness-aware models, fairness has been formalized based on well-known *notions* of justice and equity at a group level, like female vs. male or nonpopular vs. popular experts. OpeNMF2 has integrated two well-known notions:

**Demographic Parity [11]**, which requires the top- $k$  recommended experts to reflect  $p$  as the desired proportion of disadvantaged experts, *irrespective* of an expert having the required skills. For instance, given  $p = 0.5$  as the desired proportion of females, a debiasing reranker should preserve the 1:1 balance in the top- $k$  ranked list of recommended experts between female and male experts. This notion of fairness forego qualifications and has limitations [11, 23]. Notably, a random baseline that assigns experts to teams from a uniform distribution of experts regardless of popularity or gender labels is an *ideally* fair model yet at the cost of very low success rates for the recommended teams;

**Equal Opportunity [23]**, which enforces the desired proportion of disadvantaged experts only among the *qualified* experts. For example, given ‘*machine learning*’ as the required skill and  $p = 0.5$ , a debiasing reranker should select *only* among experts who have ‘*machine learning*’ in their skills while preserving the 1:1 balance between female and male experts in the top- $k$  ranked experts.

**3.1.3 Fairness Metrics.** OpeNMF2 integrated fairness evaluation metrics, including skew and normalized discounted cumulative KL-divergence (ndkl) [16]. While skew calculates the logarithmic ratio of the proportion of disadvantaged experts in top- $k$  list of experts to the desired proportion, ndkl measures the divergence of the distributions using Kullback–Leibler [36]. In both metrics, the lower, the better, with being 0 in the ideal equal distributions. In tandem with accuracy metrics such as precision, recall, ndcg, and map at top- $k$ , which measure the efficacy of the model at recommending *correct* experts for the teams in a test set, OpeNMF2 allows exploring the trade-offs between notions of fairness and the model’s accuracy for a team formation model.

**3.1.4 Gender vs. Popularity Labels.** OpeNMF2 relies on experts’ labels of popularity or gender to measure the biases before and after running the debiasing rerankers. Since gender is self-identified, gender labels are either available, like in uspt dataset, or have been inferred, like in imdb, based on the roles identified as actors or actresses. For the popularity status, an expert has been *objectively* labelled based on the number of teams the expert has participated in, referred to as *sociometric* popularity, which is widely adopted in recommender system literature [35]. As shown earlier in Figure 1 (bottom), we adopt two alternatives in OpeNMF2: *i*) avg: an expert is popular if the expert participated in more than the average number of teams per expert over the entire dataset, and nonpopular otherwise, or *ii*) auc: an expert is popular if she belongs to the *short head* in the 2-d curve of the distribution of experts in teams, and nonpopular otherwise. We split the curve into a short head and a long tail based on equal area under the curve. We acknowledge that an expert’s participation in many teams, e.g., movies in imdb, does not necessarily reflect popularity from the people’s perspectives (i.e., being liked), but repetition of the expert in many training samples

**Table 2: Results of debiasing rerankers on the Bayesian (variational) neural classifier (Bnn), with skill embeddings (–emb) and lack thereof, for mitigating popularity bias on imdb.**

|                    | neural model | reranker         |       | %ndkl10↓ |              | %map10↑     |   | %ndcg10↑    |   |
|--------------------|--------------|------------------|-------|----------|--------------|-------------|---|-------------|---|
|                    |              |                  |       | before   | after        | Δ           | Δ | Δ           | Δ |
| demographic parity | Bnn [44]     | det-cons [16]    | 67.53 |          | <b>16.59</b> | -0.36       |   | -0.82       |   |
|                    |              | det-greedy [16]  |       |          | <b>16.60</b> | -0.36       |   | -0.82       |   |
|                    |              | det-relaxed [16] |       |          | <b>16.35</b> | -0.36       |   | -0.82       |   |
|                    |              | fa-ir [63]       |       |          | 17.27        | <b>0.00</b> |   | <b>0.00</b> |   |
|                    | Bnn-emb [44] | det-cons [16]    | 74.67 |          | 15.71        | -0.48       |   | -1.03       |   |
|                    |              | det-greedy [16]  |       |          | 15.72        | -0.48       |   | -1.03       |   |
|                    |              | det-relaxed [16] |       |          | <b>15.43</b> | -0.48       |   | -1.03       |   |
|                    |              | fa-ir [63]       |       |          | <b>17.53</b> | <b>0.00</b> |   | <b>0.00</b> |   |
| equal opportunity  | Bnn [44]     | det-cons [16]    | 61.74 |          | <b>19.85</b> | -0.35       |   | -0.81       |   |
|                    |              | det-greedy [16]  |       |          | <b>20.11</b> | -0.35       |   | -0.81       |   |
|                    |              | det-relaxed [16] |       |          | <b>19.70</b> | -0.35       |   | -0.81       |   |
|                    |              | fa-ir [63]       |       |          | <b>16.61</b> | <b>0.00</b> |   | <b>0.00</b> |   |
|                    | Bnn-emb [44] | det-cons [16]    | 68.57 |          | <b>18.94</b> | -0.48       |   | -1.03       |   |
|                    |              | det-greedy [16]  |       |          | <b>19.21</b> | -0.48       |   | -1.03       |   |
|                    |              | det-relaxed [16] |       |          | <b>18.77</b> | -0.48       |   | -1.03       |   |
|                    |              | fa-ir [63]       |       |          | <b>16.88</b> | <b>0.00</b> |   | <b>0.00</b> |   |

of teams from the neural model’s perspective does and leads to the model’s popularity bias.

**3.1.5 Case Studies of Fairness.** OpeNMF2’s case studies of debiasing algorithms on imdb to mitigate popularity bias based on the two notions of fairness have partially shown in Table 2. As seen, deterministic rerankers (det-\*) [16] reduce the divergence with the desired distribution and mitigate popularity bias, as shown by lower ndkl, but at the cost of negative impacts on models’ accuracy. Similar trends can be observed in dblp dataset [39]. However, the probabilistic reranker (fa-ir) [63] could mitigate the bias while maintaining the models’ accuracy. For the complete comparative results, we refer the reader to our fairness-related papers [39, 40].

## 3.2 Graph Neural Team Formation

The use of experts’ collaboration graphs was mainstream in traditional team formation approaches rooted in subgraph optimization [17, 31, 50]. However, recent neural models have increasingly leveraged them for pretraining and *transfer* learning of dense vector representations of skills, enabling scalability to a large set of skills and reducing the complexity of neural classifiers in their input layers [32, 42, 43], as illustrated in Figure 4 (left). Such approaches entail two key shortcomings: *i*) the skill embeddings are learned disjointly in a self-supervised manner during a pretraining phase, oblivious to the team formation learning process, and *ii*) they suffer from the curse of sparsity in the output layer of the neural classifiers. To fill the gaps, from Figure 4 (right), we reformulate the team formation problem into *end-to-end* link predictions between nodes of required skills and optimal experts for a team node directly within the expert collaboration graph. The end-to-end formulation directly taps into multi-hop relational and structural information in the graph for complex cross-team links among experts and their associated skills, as shown in Figure 5. Moreover, it avoids two disjoint learning processes with two sets of parameters; that is, pretraining of skill vectors and fine-tuning for multi-label classification, hence excelling in computational efficiency and resource usage. Lastly, it alleviates the curse of sparsity as we transition from Boolean



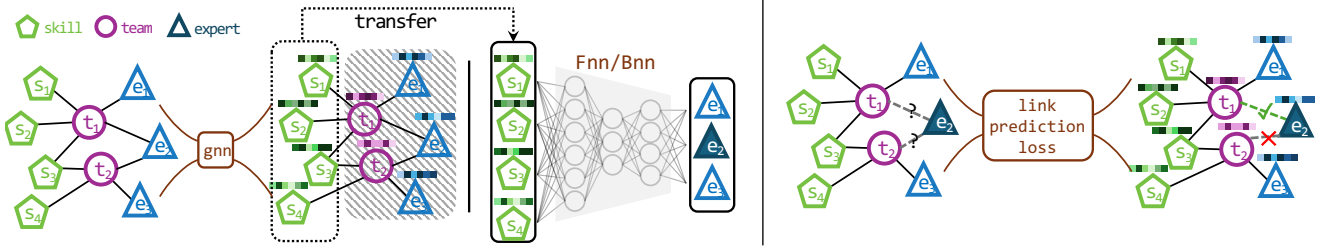


Figure 4: Transfer-based graph neural network [1] (left) versus end-to-end approach [2] (right).

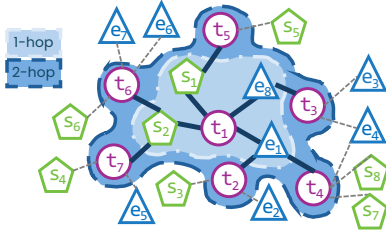


Figure 5: Multi-hop relational and structural information in the expert collaboration graph.

multi-label classification to link predictions between expert nodes and a test team node linked to its required skill nodes. Here, we provide an overview of the principal components; for theoretical details, we refer to graph-related papers [1, 2, 43].

**3.2.1 Expert Collaboration Graph.** As shown in Figure 4, given a successful team  $(s, e)_{y=1} \in \mathcal{T}$ , we map it into a connected *star* subgraph with a team node as the central node and its required skills  $s$  and expert members  $e$  are the leaves, connected to the team node but not to each other. After incorporating all successful teams of a training set in a heterogeneous graph  $G = \langle N, \mathcal{L} \rangle$ , whose nodes are skills  $n_S$ , successful teams  $n_T$  and experts  $n_E$ , and links are  $\mathcal{L} = \{n_S \times n_T\} \cup \{n_T \times n_E\}$ , we then train a graph neural network  $g$ , parameterized by  $\phi, g_\phi : \mathcal{N} \rightarrow \mathbb{R}^d$ , via link prediction loss to learn dense vectors of skills and experts using positive (existing) and negative (non-existing) link sampling strategies. To study the impact of graph structure, OpenTF2 currently has a variety of graph structures, including skill-member bipartite, skill-team-member tripartite, and skill-team-member-**location** when the geolocation information for the teams is available, like in dblp.

**3.2.2 Transfer vs. End-to-End Learning.** Via PyG<sup>2</sup> [13], we now have integrated a diverse set of graph neural networks for  $g_\phi$ , including random-walk-, metapath-, message-passing-, and attention-based models, such as node2vec (n2v) [20], metapath2vec (m2v) [10], graph convolutional network (gcn) [62], graphsage (gs) [21], graph attention network (gat) [3], and graph isomorphic network (gin) [59]. In the transfer-based setting, as shown in Figure 4 (left), OpenTF2 operates in the embedding mode, where a graph neural network is trained to learn skill embeddings while disregarding embeddings of other node types. These skill embeddings are then fed into an underlying multi-label classifier, such as a non-variational

Table 3: Efficacy of end-to-end (e2e) vs. transfer (t-\*) approaches in dblp and imdb using skill-team-expert graph.

|       |     | dblp         |              |              |              | imdb         |              |              |              |
|-------|-----|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|       |     | %ndcg5       | %ndcg10      | %map5        | %map10       | %ndcg5       | %ndcg10      | %map5        | %map10       |
| e2e   | gs  | 33.21        | 38.39        | 25.18        | 28.10        | <b>33.70</b> | 36.78        | 27.00        | 29.03        |
|       | gin | 31.50        | 36.12        | 22.74        | 25.36        | 28.49        | 32.79        | 21.76        | 24.30        |
|       | gat | <b>38.11</b> | <b>43.78</b> | <b>31.44</b> | <b>34.65</b> | 33.45        | <b>37.40</b> | <b>27.17</b> | <b>29.68</b> |
| t-Fnn | gs  | 01.26        | 01.63        | 00.60        | 00.76        | 00.95        | 01.27        | 00.44        | 00.55        |
|       | gin | 01.37        | 01.57        | 00.59        | 00.76        | 00.87        | 01.19        | 00.40        | 00.51        |
|       | gat | 01.62        | 02.08        | 00.92        | 01.09        | 01.06        | 01.43        | 00.50        | 00.62        |
| t-Bnn | gs  | 00.94        | 01.36        | 00.52        | 00.65        | 01.12        | 01.55        | 00.54        | 00.67        |
|       | gin | 00.84        | 01.15        | 00.49        | 00.60        | 01.07        | 01.47        | 00.52        | 00.65        |
|       | gat | 00.87        | 01.28        | 00.49        | 00.62        | 01.10        | 01.50        | 00.53        | 00.66        |

feedforward neural network (Fnn) or a variational Bayesian neural network (Bnn). During inference, given a subset of skills  $s$  required to form an unseen test team, the corresponding skill embeddings are summed and fed into the neural classifier per Equation 1, i.e.,  $f_\theta(s) \simeq f_\theta(v_s = \sum_{s_i \in s} g_\phi(n_{s_i}))$ , and the experts with top- $k$  highest probabilities are selected as the recommended team of size  $k$ . In contrast, in the end-to-end scenario, a graph neural network is trained to directly predict member-team links, using them as supervision signals. During inference, as shown in Figure 4 (right), given a test team with its subset of required skills yet *unseen* expert members, we use the graph neural network to predict links between expert nodes  $\mathcal{E}$  and the team's node  $t$  as cosine similarity between their embeddings. Formally,  $\forall e_j \in \mathcal{E}; f_\theta(s) \simeq f(s : G, g_\phi) = g_\phi(n_{e_j})^\top g_\phi(n_t)$  in  $G(\mathcal{N}, \mathcal{L})$ . As seen, the end-to-end approach has a single set of parameters  $\phi$ , eliminating  $\theta$  for the multi-label classifier, and hence, excels in computational efficiency and resource usage. For sample runs in *transfer* or *end-to-end* settings, see our colab script at the codebase.<sup>3</sup>

**3.2.3 Case Studies of Graph-based Learning.** From Table 3, we observe that the end-to-end approach (e2e) consistently and substantially outperforms *all* transfer-based methods using either non-variational (t-Fnn) or variational Bayesian (t-Bnn) classifiers across all employed graph neural networks, datasets, and metrics. The superiority of the end-to-end approach lies in leveraging both multi-hop relational and structural information in the expert collaboration graph and supervised information about the optimal subset of experts within teams. For the full comparative results, we refer to our paper currently under review [2].

<sup>2</sup><https://pyg.org/><sup>3</sup><https://github.com/fani-lab/OpenTF/blob/main/ipybn/gnn.ipynb>

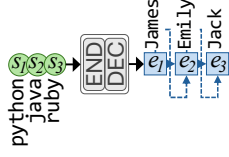


Figure 6: Translative team formation [53].

Table 4: Efficacy of translative vs. multi-label models in dblp.

|                   | $k$ | transformer [54] | conv2s [15] | rnn-att [65]  | Bnn    | Bnn-emb |
|-------------------|-----|------------------|-------------|---------------|--------|---------|
| Row 1: %precision | 2   | <b>10.4119</b>   | 2.4998      | <u>3.6176</u> | 0.0570 | 0.1124  |
|                   | 5   | <b>7.0113</b>    | 1.6122      | <u>2.3581</u> | 0.0663 | 0.1290  |
|                   | 10  | <b>3.5392</b>    | 0.8242      | <u>1.1992</u> | 0.0710 | 0.1251  |
| %recall           | 2   | <b>6.3457</b>    | 1.5071      | <u>2.1698</u> | 0.0351 | 0.0668  |
|                   | 5   | <b>10.5477</b>   | 2.4177      | <u>3.5115</u> | 0.0993 | 0.1909  |
|                   | 10  | <b>10.6397</b>   | 2.4760      | <u>3.5753</u> | 0.2118 | 0.3699  |
| %ndcg             | 2   | <b>10.3611</b>   | 2.4770      | <u>3.5822</u> | 0.0538 | 0.1083  |
|                   | 5   | <b>10.4597</b>   | 2.4276      | <u>3.5184</u> | 0.0806 | 0.1555  |
|                   | 10  | <b>10.4824</b>   | 2.4487      | <u>3.5391</u> | 0.1330 | 0.2397  |
| %map              | 2   | <b>5.9463</b>    | 1.3554      | <u>1.9412</u> | 0.0242 | 0.0474  |
|                   | 5   | <b>9.2909</b>    | 2.0008      | <u>2.8791</u> | 0.0411 | 0.0792  |
|                   | 10  | <b>9.3210</b>    | 2.0127      | <u>2.8930</u> | 0.0558 | 0.1033  |

### 3.3 Translative Team Formation

From Figure 6, the team formation problem can be viewed as a translation task, mapping a dynamic-length input sequence of required skills onto a dynamic-length output sequence of experts while leveraging the autoregression and global attention mechanisms, which capture dependencies beyond independent expert probabilities in multi-label classification. In OpenNTF2, we integrated OpenNMT-py<sup>4</sup> [34] to utilize modern transformers and encoder-decoder recurrent models with attention mechanisms [5, 28, 29, 41, 52]. Formally, we estimate  $f_{\theta}(\mathbf{s})$  in Equation 1 on a parallel dataset whose pairs of sequences are the ordered list of required skills  $\mathbf{s} = [s_1, \dots, s_n]$  into an optimum ordered list of experts  $\mathbf{e} = [e_1, \dots, e_m]$  for each successful team  $(\mathbf{s}, \mathbf{e})_{y=1} \in \mathcal{T}$ . An encoder maps the sequence of skills  $[s_1, \dots, s_n]$  onto  $\mathbf{h}_n$  and a decoder generates the sequence of experts  $[e_1, \dots, e_m]$  from the  $\mathbf{h}_n$ , one expert at a time, decomposing the conditional probability  $p(\mathbf{e}|\mathbf{s}, \theta)$  as  $\prod_{k=1}^m p(e_k|e_{<k}, \mathbf{s}, \theta)$  and seeking the maximum probability among subsets of experts as an optimum team for  $\mathbf{s}$ , i.e.,  $f_{\theta}(\mathbf{s}) = \mathbf{e}$ . The probability of generating an expert at the decoder can be conditioned not only on  $\mathbf{h}_n$  but also on all  $\mathbf{h}_{<n}$  at the encoder, enabling the decoder to *attend* to all skills in the input sequence selectively [41]. To reduce the computational complexity at the encoder and the decoder, a model may have *no* recurrent connections, like in transformers [54], enabling parallel calculation of  $\mathbf{h}_{<n}$  at the encoder and  $\mathbf{h}_{\geq n}$  at the decoder, an architecture that yielded promising performance on machine translation and led to extensive research on sequence modelling [9, 48, 61]. For sample runs in encoder-decoder settings, see our colab script at the codebase.<sup>5</sup>

**3.3.1 Case Studies of Translative Learning.** From Table 4, translative models yield better performance vs. multi-label neural classifiers (Bnn-\*) across metrics in dblp dataset, which can be

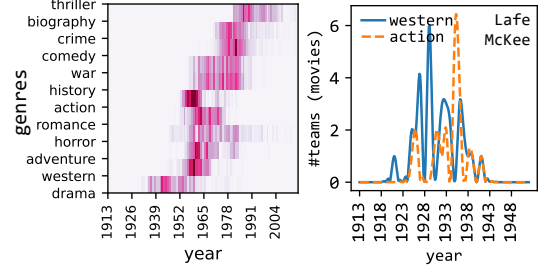


Figure 7: Temporal distribution of movies over genres (left), and temporal participation of an actor in two genres (right).

attributed to conditioning of expert recommendation on the previously recommended experts in the output sequence, addressing the inherent sparse activations in the output layer of neural classifiers. For the complete comparative results, we refer the reader to our paper [53].

### 3.4 Temporal Training Strategy

As time elapses, skill proficiency or interests of experts evolve due to the shift in societal needs, emerging technologies, and working experiences. Therefore, time as an aspect should be considered in models to incorporate the possible drift and variability of experts' skills and collaboration ties. A collaboration between experts that was successful in the past may not necessarily guarantee success in the future. For instance, Figure 7 demonstrates the *non*-uniform temporal distribution of movies over genres (skills) (left) and temporal activity of an actor toward two movie genres (right) within *yearly* time intervals in imdb. Incorporation of time has already shown performance improvement in other literature, such as information retrieval [22], knowledge graphs [18, 38], and recommender systems [26]. However, prior team formation methods learn experts and skills vector representation in a *static* latent space [7, 43, 47, 49], overlooking the temporality of experts' skills in time and its impact on the recommendation of future teams.

Inspired by curriculum learning [58] and temporal latent space inference [66], we developed a temporal training strategy [12] to consume time information as an aspect of experts' skills and collaboration ties in teams, as opposed to an extra numeric input feature, while being model-agnostic and with *no* modification to the models' architecture<sup>6</sup>. In contrast to the independent and identically distributed (i.i.d) assumption (*bag of teams*) during model training on a shuffled dataset, we order the teams based on time intervals (e.g., yearly) and train a neural model incrementally on streamed subsets of teams. From Figure 8, we randomly initialize the model's parameters at  $t=0$  and train it on the subset of teams in the first time interval  $\mathcal{T}_{t=1}$  for a number of epochs. We then continue training on the subsequent time intervals, using the learned parameters from the previous time interval. This process is repeated until we complete the training on the subset of teams in the last time interval  $\mathcal{T}_{t=T}$ . Temporal training moves skill and expert embeddings in their optimal positions in the latent space for recommending experts

<sup>4</sup><https://github.com/OpenNMT/OpenNMT-py>

<sup>5</sup><https://github.com/fani-lab/OpenNTF/blob/main/ipynb/nmt.ipynb>

<sup>6</sup>The temporal training strategy is compatible with variational (Bnn) and non-variational neural classifiers (Fnn). It does not apply to seq-to-seq models and transformers. Support for temporal graph neural networks is in the future release.

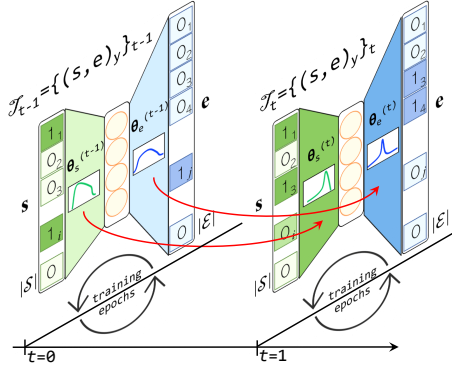


Figure 8: Temporal training strategy [12].

for a team at the future time interval  $t=T+1$ . For sample temporal training of models, see our colab script at the codebase.<sup>7</sup>

**3.4.1 Case Studies of Temporal Learning.** Table 5, which shows partial results for Bayesian models on dblp and imdb datasets, along with our comprehensive empirical results [12] demonstrate that neural models that utilize the temporal training strategy (tBnn and tBnn-emb) excel at the efficacy of recommended experts for teams. This is in contrast to existing neural models, which make the simplifying independent and identically distributed (i.i.d) assumption in training instances of teams that discounts models' efficacy due to possible drift of experts' skills and collaboration ties over time.

From Figure 2, we developed tNtf class, which inherits from the Ntf class to override the standard methods (interface) for model training, testing, and evaluation steps. However, contrary to neural models, tNtf is a proxy (placeholder) class that contains a neural model and brings it to the temporal training flow.

### 3.5 Enhanced Benchmarks and Software Design

In OpenTF2, we also contribute github dataset of open-source repositories as teams of software developers crawled from github<sup>8</sup>, including contributors as the experts and programming languages as the skills as well as complementary information such as stargazer count, number of forks, and creation date. OpenTF2 additionally introduces new evaluation metrics, including skill coverage [44, 55], to capture the coverage of required skills in the recommended team. From software engineering, OpenTF2 adopts a component-based object-oriented design in which components are loosely coupled and reusable. Dependencies are installed on demand at the component level to improve the efficiency of the execution environment and prevent package conflicts, which is essential when integrating multiple models, training strategies, and evaluation components within a unified framework. OpenTF2 further incorporates continuous integration and testing<sup>9</sup>, along with containerized execution, to ensure consistent and reproducible behavior across environments. Lastly, configurations are handled via hierarchical

<sup>7</sup><https://github.com/fani-lab/OpenTF/blob/main/ipynb/tntf.ipynb>

<sup>8</sup><https://github.com/>

<sup>9</sup><https://github.com/fani-lab/OpenTF/tree/main/.github/workflows>

Table 5: Performance improvement of temporal training.

|      | neural model | %pr2          | %pr5          | %pr10         | %ndcg2        | %ndcg5        | %ndcg10       | %aucroc      |
|------|--------------|---------------|---------------|---------------|---------------|---------------|---------------|--------------|
| dblp | Bnn          | 0.0570        | 0.0663        | 0.0710        | 0.0538        | 0.0806        | 0.1330        | 63.52        |
|      | Bnn-emb [44] | 0.1124        | 0.1290        | 0.1251        | 0.1083        | 0.1555        | 0.2397        | 66.81        |
|      | tBnn         | 0.1189        | 0.1413        | 0.1664        | 0.1126        | 0.1689        | 0.3031        | 73.08        |
|      | tBnn-emb     | <b>0.2996</b> | <b>0.2938</b> | <b>0.2811</b> | <b>0.3048</b> | <b>0.3860</b> | <b>0.5721</b> | <b>74.83</b> |
| imdb | Bnn          | 0.2128        | 0.5106        | 0.4255        | 0.1646        | 0.5699        | 0.7848        | 51.16        |
|      | Bnn-emb [44] | 0.4255        | 0.5106        | 0.6383        | 0.3292        | 0.5923        | 1.1358        | 51.82        |
|      | tBnn         | <b>0.8511</b> | <b>1.5319</b> | <b>1.4043</b> | <b>0.7548</b> | <b>1.7381</b> | <b>2.6829</b> | <b>63.46</b> |
|      | tBnn-emb     | <b>0.8511</b> | <b>1.1064</b> | <b>1.0638</b> | <b>0.9474</b> | <b>1.4848</b> | <b>2.2007</b> | <b>66.87</b> |

hydra [60]<sup>10</sup>, enabling systematic experiments and extensive ablation studies.

## 4 Quick Start

Below is an example command with hydra config overrides to adjust the default settings of each component as needed:<sup>11</sup>

```
python -u ./src/main.py "cmd=[prep, train, test, eval, fair]"

"models.instances=[mdl.bnn.Bnn,                               #Bayesian classifier
                    mdl.nmt.Nmt,                               #translative model
                    mdl.emb.gnn.Gnn,                           #graph-based (e2e)
                    mdl.tntf.tntf_mdl.bnn.Bnn]"               #temporal training of Bnn

"data.domain=cmn.publication.Publication"
"data.source=./data/dblp/toy.dblp.v12.json"
"data.output=./output/dblp/toy.dblp.v12.json"
"data.filter.min_team_size=5"
"data.filter.min_nteam=10"                                     #number of teams per expert
"data.embedding.class_method=mdl.emb.gnn.Gnn_gs"              #GraphSAGE (e2e, transfer)

"train.nfolds=3"
"train.train_test_ratio=0.85"
"train.save_per_epoch=3"
"train.step_ahead=1"

"test.per_epoch=True"
"test.topK=1000"                                               #keep topK probs only

"eval.metrics.topk='2,5,10'"
"eval.metrics.trec=[ndcg_cut_topk, map_cut_topk]"
"eval.metrics.fair=[ndkl, skew]"
"eval.metrics.other=[skill_coverage_topk]"
"eval.per_instance=True"                                       #for paired sig. tests

"fair.fgender=./data/dblp/toy.dblp.v12.json.females.csv"
"fair.algorithm=[fa-ir, det-greedy]"
"fair.notion=[eo, dp]"
"fair.attribute=[gender, popularity]"
"fair.is_popular_alg=[avg]"
```

## 5 Concluding Remarks and Future Work

We presented OpenTF2 with key extensions to its initial release including *i*) fairness-aware rerankers to mitigate popularity and gender disparities; *ii*) graph neural network for transferring embeddings of skills as well as end-to-end team formation via link prediction; *iii*) encoder-decoder recurrent models and transformers for team formation framed as a sequence prediction (translation) task; *iv*) and training strategy to incorporate temporal aspects of experts' skills and collaboration ties within time. OpenTF2's future focus is on fairness, incorporating in-process and pre-process mitigation methods, as well as temporal graph neural networks.

<sup>10</sup><https://hydra.cc/>

<sup>11</sup><https://github.com/fani-lab/OpenTF/blob/main/ipynb/quickstart.ipynb>

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